theory *Week_1_sol*
imports *Main*
begin

King's College London
Department of Informaitcs

Semester 1

Formal Verification: 6CCS3VER

Exercise Sheet 6

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called Week_1.thy and add the following to its beginning.

theory Week_1
imports Main
begin

Exercise 6.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

```
lemma
  assumes T: "T b"
      and A: "A a ∧ A b ⟹ a = b"
      and TA: "⋀x. T x ⟹ A x"
      and Aab: "A a"
  shows "T a"

proof-

  have 1: "A b"
    using TA[OF T]
    .
```

```
  have 2: "A a ∧ A b"
    apply(rule conjI)
    using Aab 1
    .

  have 3: "a = b"
    apply(rule A)
    using 2
    .

  show ?thesis
    using T
    apply(subst 3)
    .
qed

end
```